

- ? In what ways does a methyl group differ chemically from the other six important chemical groups shown in Figure 4.9?

TEST YOUR UNDERSTANDING

➔ For more multiple-choice questions, go to the **Practice Test** in the **Mastering Biology** eText or Study Area, or go to goo.gl/GruWRg.

Levels 1-2: Remembering/Understanding

- Organic chemistry is currently defined as
 - the study of compounds made only by living cells.
 - the study of carbon compounds.
 - the study of natural (as opposed to synthetic) compounds.
 - the study of hydrocarbons.
- VISUAL SKILLS** Which functional group is present in this molecule?

- sulfhydryl
 - carboxyl
 - methyl
 - phosphate
- MAKE CONNECTIONS** Which chemical group is most likely to be responsible for an organic molecule behaving as a base (see Concept 3.3)?

- hydroxyl
 - carbonyl
 - amino
 - phosphate

Levels 3-4: Applying/Analyzing

- VISUAL SKILLS** Visualize the structural formula of each of the following hydrocarbons. Which hydrocarbon has a double bond in its carbon skeleton?

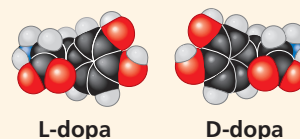
- C_3H_8
 - C_2H_6
 - C_2H_4
 - C_2H_2
- VISUAL SKILLS** Choose the term that correctly describes the relationship between these two sugar molecules.

 - structural isomers
 - cis-trans* isomers
 - enantiomers
 - isotopes
- VISUAL SKILLS** Identify the asymmetric carbon in this molecule.
- Which action could produce a carbonyl group?
 - the replacement of the $-OH$ of a carboxyl group with hydrogen
 - the addition of a thiol to a hydroxyl
 - the addition of a hydroxyl to a phosphate
 - the replacement of the nitrogen of an amine with oxygen

- VISUAL SKILLS** Which of the molecules shown in question 5 has an asymmetric carbon? Which carbon is asymmetric?

Levels 5-6: Evaluating/Creating

- EVOLUTION CONNECTION • DRAW IT** Some scientists think that life elsewhere in the universe might be based on the element silicon, rather than on carbon, as on Earth. Look at the electron distribution diagram for silicon in Figure 2.7 and draw the Lewis dot structure for silicon. What properties does silicon share with carbon that would make silicon-based life more likely than, say, neon-based life or aluminum-based life?
- SCIENTIFIC INQUIRY** Fifty years ago, pregnant women who were prescribed thalidomide for morning sickness gave birth to children with birth defects. Thalidomide is a mixture of two enantiomers; one reduces morning sickness, but the other causes severe birth defects. Today, the FDA has approved this drug for non-pregnant individuals with Hansen's disease (leprosy) or newly diagnosed multiple myeloma, a blood and bone marrow cancer. The beneficial enantiomer can be synthesized and given to patients, but over time, both the beneficial *and* the harmful enantiomer can be detected in the body. Propose a possible explanation for the presence of the harmful enantiomer.
- WRITE ABOUT A THEME: ORGANIZATION** In 1918, an epidemic of sleeping sickness caused an unusual rigid paralysis in some survivors, similar to symptoms of advanced Parkinson's disease. Years later, L-dopa (below, left), a chemical used to treat Parkinson's disease, was given to some of these patients. L-dopa was remarkably effective at eliminating the paralysis, at least temporarily. However, its enantiomer, D-dopa (right), was subsequently shown to have no effect at all, as is the case for Parkinson's disease. In a short essay (100–150 words), discuss how the effectiveness of one enantiomer and not the other illustrates the theme of structure and function.



12. SYNTHESIZE YOUR KNOWLEDGE



Explain how the chemical structure of the carbon atom accounts for the differences between the male and female lions seen in the photo.

For selected answers, see Appendix A.